

Nour Amgad Elhwary

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EDUCATION

Misr International University

Bachelor Degree in Computer Science

Cairo, Egypt

Oct. 2022 – Jun. 2026

EXPERIENCE

PwC (Etic by PwC)

Data Science Intern

Cairo, Egypt

Jun. 2025 – Sep. 2025

- Developed an AI powered CV analysis and decision support tool using RAG, LangGraph, and large language models to deliver real time candidate insights.

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Data Science & AI Intern

Cairo, Egypt

Jul. 2024 – Sep. 2024

- Selected for a two month extension following an initial one month internship, utilizing SQL and Teradata to retrieve and analyze enterprise datasets for data analysis tasks.
- Performed data cleaning, exploratory data analysis, and predictive analytics projects to identify trends and support data driven decision making.

Misr International University

Co Assistant

Cairo, Egypt

Sep. 2024 – Dec. 2024

- Teaching assistant for “Introduction to Computer Science” course.

PROJECTS

ReVerie – AI Powered VR Memory Reconstruction Platform | Python, Computer Vision, Deep Learning, GCP, Supabase, Unreal Engine, Flutter

- Co-developed an AI-powered VR platform transforming personal photographs into immersive reminiscence experiences for Alzheimer’s patients.
- Contributed to memory reconstruction workflows using depth estimation, segmentation, inpainting, and 3D reconstruction of humans, objects, and environments.
- Built the Flutter mobile application end to end using Riverpod, Supabase, and Agora Video SDK, integrating AI pipeline outputs and backend APIs with the Unreal Engine VR system to support patient doctor interactions
- Developed GCP based AI pipelines integrated with Supabase to support scalable photo to VR memory reconstruction workflows.

AI-Powered Recruitment Screening System – PwC AI Studio | Python, LLMs, RAG, LangGraph, NLP

- Developed an AI powered recruitment screening system using LangGraph, Retrieval Augmented Generation (RAG), and advanced prompt engineering to analyze job descriptions and evaluate candidate resumes.
- Implemented candidate ranking and qualification extraction workflows to identify the most relevant applicants and highlight key skills and experience.
- Delivered context aware hiring insights that reduced manual screening effort and improved candidate job matching accuracy.

Smart Bin Waste Management System | DenseNet121, OpenCV, A* Search, Arduino

- Built a two agent waste management prototype combining waste classification, fill level monitoring, overflow prediction, and route optimization.
- Fine tuned DenseNet121 on a 2,527 image, 4 class dataset, achieving 98% test accuracy and 97.89% macro F1.

Multi-Label Emotion Intelligence System | PyTorch, Hugging Face, RoBERTa/BERT

- Fine tuned a transformer-based multi-label emotion classifier using class-weighted BCE loss, threshold tuning, and sarcasm aware interpretation.
- Achieved 0.598 micro F1, 0.964 hamming accuracy, and 0.555 sample Jaccard on GoEmotions; trained a sarcasm classifier with 0.635 F1.

Aston Martin Website | HTML, CSS, JavaScript, Node.js, MongoDB

- Built a car customization website with user/product database, vehicle configuration, inquiries, and selection management.

University Website – Won Best Project | C++, SFML, TGUI

- Built an instructor webpage with auth based sign in, course display, schedules, and real time sync to student views.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, PHP, Dart, SQL

Web & Mobile: Flutter, React.js, Node.js, Express.js, Bootstrap

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, 3D Reconstruction, RAG, LangGraph, LangChain, LLM Applications, Prompt Engineering

Databases: SQL Databases, Relational Databases, MongoDB, Supabase

Software: Data Structures, OOP, Algorithms, Git Version Control, Problem Solving

Hardware: Verilog Design, Digital Logic Design

Tools: Git, GitHub, Microsoft Office, Cisco Packet Tracer, Blender

RESEARCH

Emotion Detection Using NLP — IEEE Published Research

- Co-authored IEEE-published research on text based emotion detection using 6 ML models and 4 deep learning architectures.
- Conducted comparative analysis using accuracy, precision, recall, and F1 score to identify effective emotion recognition models.

ReVerie Systematic Literature Review

- Reviewed AI driven memory reconstruction, VR therapy, computer vision, digital twins, and cognitive stimulation technologies.

Optimizing Network Flow Using Edmonds Karp Algorithm

- Selected as one of the top projects during Scientific Day, with potential for future publication.

LANGUAGES

Languages: English (Fluent), Arabic (Fluent), German (Intermediate)